

Lab 3: Monitoring, Logging & DevSecOps

Provision scalable infrastructure

Provision scalable infrastructure using Terraform (Infrastructure as Code): write configuration files, initialize, plan, and apply the infrastructure.

```
Last login: Fri Oct 2 01:15:30 on ttys000
anushkaberlia@Anushkas-MacBook-Air-2 ~ % docker version
docker ps
Client:
  Version:           20.7.2
  API version:       1.55
  Go version:        go1.26.5
  Git commit:        a7dcaab
  Built:             Wed Aug 5 18:27:50 2026
  OS/Arch:           darwin/arm64
  Context:           desktop-linux

Server: Docker Desktop 4.98.0 (238679)
Engine:
  Version:          20.7.2
  API version:      1.55 (minimum version 1.48)
  Go version:       go1.26.5
  Git commit:       4a9b3d1
  Built:            Wed Aug 5 18:28:35 2026
  OS/Arch:          linux/arm64
  Experimental:     false
  containerd:
    Version:        v2.3.3
    GitCommit:      aad11806b69517fcd3809450b6f82de282e19b
  runc:
    Version:        1.4.3
    GitCommit:      v1.4.3-0-gbb14dabe
  docker-init:
    Version:        0.19.0
    GitCommit:      da48ada0

CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
85854952a6b4   0988e772b9f9   "/docker-entrypoint..." 10 hours ago   Up 10 hours   127.0.0.1:8092->80/tcp    tf-web-2
8306d3b9ce7e   0988e772b9f9   "/docker-entrypoint..." 10 hours ago   Up 10 hours   127.0.0.1:8091->80/tcp    tf-web-1
6e17d3498623   ansible-ubuntu-lab:1 "sleep infinity"         11 hours ago   Up 11 hours   127.0.0.1:9882->80/tcp    ansible-node2
39ab384586f6   ansible-ubuntu-lab:1 "sleep infinity"         11 hours ago   Up 11 hours   127.0.0.1:8081->80/tcp    ansible-node1

anushkaberlia@Anushkas-MacBook-Air-2 ~ % docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
85854952a6b4   0988e772b9f9   "/docker-entrypoint..." 10 hours ago   Up 10 hours   127.0.0.1:8092->80/tcp    tf-web-2
8306d3b9ce7e   0988e772b9f9   "/docker-entrypoint..." 10 hours ago   Up 10 hours   127.0.0.1:8091->80/tcp    tf-web-1
6e17d3498623   ansible-ubuntu-lab:1 "sleep infinity"         11 hours ago   Up 11 hours   127.0.0.1:9882->80/tcp    ansible-node2
39ab384586f6   ansible-ubuntu-lab:1 "sleep infinity"         11 hours ago   Up 11 hours   127.0.0.1:8081->80/tcp    ansible-node1

anushkaberlia@Anushkas-MacBook-Air-2 ~ % ls -l /var/run/docker.sock
ls: /var/run/docker.sock: No such file or directory
anushkaberlia@Anushkas-MacBook-Air-2 ~ % docker context inspect --format '{{.Endpoints.docker.Host}}'
unix:///Users/anushkaberlia/.docker/run/docker.sock
anushkaberlia@Anushkas-MacBook-Air-2 ~ % ls -l /Users/anushkaberlia/.docker/run/docker.sock
-rwxr-xr-x@ 1 anushkaberlia  staff  0  1 Oct 21:21 /Users/anushkaberlia/.docker/run/docker.sock
anushkaberlia@Anushkas-MacBook-Air-2 ~ % brew tap hashicorp/tap
brew install hashicorp/tap/terraform
terraform version

mkdir -p ~/devops-two-labs/two-labs/02_terraform
cd ~/devops-two-labs/two-labs/02_terraform
[==> Downloading Homebrew API data
[==> JSON API packages.arm64.tahoe.jws.json                               Downloaded 15.6MB/15.6MB
[==> Auto-updating Homebrew...
Adjust how often this is run with 'SHCMBREW_AUTO_UPDATE_SECS' or disable with
'SHCMBREW_NO_AUTO_UPDATE=1'. Hide these hints with 'SHCMBREW_NO_ENV_HINTS=1' (see 'man brew').
Warning: hashicorp/tap/terraform 1.16.4 is already installed and up-to-date.
To reinstall 1.16.4, run:
```

```
To reinstall 1.16.4, run:
brew reinstall terraform
Terraform v1.16.4
on darwin_arm64
anushkaberlia@Anushkas-MacBook-Air-2 02_terraform % cat > terraform.tfvars << 'EOF'
replica_count = 2
base_port     = 8091
docker_host   = "unix:///Users/anushkaberlia/.docker/run/docker.sock"
EOF
anushkaberlia@Anushkas-MacBook-Air-2 02_terraform % ls
cat terraform.tfvars
initial.tfplan      outputs.tf           terraform.tfvars
main.tf             terraform.tfstate   variables.tf
replica_count = 2
base_port     = 8091
docker_host   = "unix:///Users/anushkaberlia/.docker/run/docker.sock"
anushkaberlia@Anushkas-MacBook-Air-2 02_terraform % terraform init
terraform fmt
terraform validate
terraform plan -out=initial.tfplan
terraform show initial.tfplan
terraform apply initial.tfplan
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of krouzwerker/docker from the dependency lock file
- Using previously-installed krouzwerker/docker v4.2.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.

docker_network.lab: Refreshing state... [id=7b0dd3eeed8179c0c8f81836e27c43e8d3efcb339b0b0a563d8f2c743f2e]
docker_image.web: Refreshing state... [id=a8a25610985e772fb9772e6fa9980da85fca509c440e8780ad3a871545afa9d2c27094mqinx:stable-alpine]
docker_container.web[0]: Refreshing state... [id=b38d3b9ce7ee88128ada71186297ecfba54fa76fb88ab6b33a6872cec94f09d]
docker_container.web[1]: Refreshing state... [id=85854952a6b4fca8e2bcd5d55cb8dcdd3524325d1b97176b18ab7f6ed2227f6]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

container_names = [
  *tf-web-1,
]
```

```

"tf-web-1",
"tf-web-2",
]
web_urls = [
"http://127.0.0.1:8091",
"http://127.0.0.1:8092",
]
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % terraform output
terraform state list
docker ps --filter name=tf-web
curl -fsS http://127.0.0.1:8091/
curl -fsS http://127.0.0.1:8092/
container_names = [
"tf-web-1",
"tf-web-2",
]
web_urls = [
"http://127.0.0.1:8091",
"http://127.0.0.1:8092",
]
docker_container.web[0]
docker_container.web[1]
docker_image.web
docker_network.lab
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS               NAMES
8d88a62a3bb6        8985e772fb9f       "/docker-entrypoint..." 10 hours ago        Up 10 hours        127.0.0.1:8092->80/tcp   tf-web-2
b3b8d3b0ce7e        8985e772fb9f       "/docker-entrypoint..." 10 hours ago        Up 10 hours        127.0.0.1:8091->80/tcp   tf-web-1
<DOCTYPE html>
<html>
<head>
<title>Welcome to nginx</title>
<style>
html { color-scheme: light dark; }
body { width: 350px; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://fs.com/nginx">fs.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
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<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://fs.com/nginx">fs.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % cat > terraform.tfvars << 'EOF'
replica_count = 4
base_port     = 8091
docker_host   = "unix:///Users/anushkaberlia/.docker/run/docker.sock"
EOF
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % >....
provider "docker" {
  host = var.docker_host
}

resource "docker_network" "lab" {
  name = "tf-classroom-net"
}

resource "docker_image" "web" {
  name      = "nginx:stable-alpine"
  keep_locally = true
}

resource "docker_container" "web" {
  count = var.replica_count
  name  = "tf-web-${count.index + 1}"
  image = docker_image.web.image_id

  networks_advanced {
    name = docker_network.lab.name
  }

  ports {
    internal = 80
    external = var.base_port + count.index
    id       = "127.0.0.1"
  }
}
EOF
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % >....
default     = 2

```

```

anushkaberlia@Anushkas-MacBook-Air-2 82 terraform N >....
default = 2
validation {
  condition = (
    var.replica_count >= 1 && var.replica_count <= 4 &&
    floor(var.replica_count) == var.replica_count
  )
  error_message = "Use a whole number from 1 to 4."
}
}

variable "base_port" {
  description = "First host port; four consecutive ports must be free"
  type        = number
  default     = 8091
  validation {
    condition = (
      var.base_port >= 1024 && var.base_port <= 65532 &&
      floor(var.base_port) == var.base_port
    )
    error_message = "Use a whole number from 1024 to 65532."
  }
}

variable "docker_host" {
  description = "Docker socket path"
  type        = string
  default     = "unix:///var/run/docker.sock"
}
EOF
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % cat > outputs.tf << 'EOF'
output "container_names" {
  value = docker_container.web[*].name
}

output "web_urls" {
  value = [
    for i in range(var.replica_count) :
    "http://127.0.0.1:${var.base_port + i}"
  ]
}
EOF
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % ls
initial.tfplan      terraform.tfstate      variables.tf
main.tf             terraform.tfstate.backup
outputs.tf           terraform.tfvars
anushkaberlia@Anushkas-MacBook-Air-2 82 terraform % terraform init
terraform init
terraform validate
terraform plan -out=initial.tfplan
terraform show initial.tfplan
terraform apply initial.tfplan
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of kreuzwerker/docker from the dependency lock file
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Terraform has been successfully initialized!

```

```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.

docker_network[1]: Refreshing state... [id=72b0dd3eeed179c8e7f1d335e07c43e6d3cfc2b339d0d43e562d5f7e743f7e]
docker_image.web: Refreshing state... [id=aha28610985e772fb9f729e0fa998da85fca5d9c4d0e870ed3d307154d8afa9d2e27094arginx:stable-alpine]
docker_container.web[0]: Refreshing state... [id=b38d3b9ce7ee88128ada7118a2b7ecfbab4fa76f088ab6e33a6872cc94f0wd]
docker_container.web[1]: Refreshing state... [id=d5854952a6b4fca8a8e2bcd5dd65cb8d4cd3524325d1b9711876ab2f6ed222f76]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# docker_container.web[2] will be created
+ resource "docker_container" "web" {
  + attach              = false
  + bridge              = (known after apply)
  + command             = (known after apply)
  + container_logs      = (known after apply)
  + container_read_refresh_timeout_milliseconds = 15000
  + entrypoint          = (known after apply)
  + env                 = (known after apply)
  + exit_code           = (known after apply)
  + hostname            = (known after apply)
  + id                  = (known after apply)
  + image               = "sh225e10980e772fb9f729e0fa998da85fca5d9c4d0e870ed3d307154d8afa9d2e27094"
  + init                = (known after apply)
  + ipc_mode            = (known after apply)
  + log_driver          = (known after apply)
  + logs                = false
  + memory_reservation  = 0
  + must_run            = true
  + name                = "tf-web-3"
  + network_data        = (known after apply)
  + network_mode        = "bridge"
  + platform            = (known after apply)
  + read_only           = false
  + remove_volumes     = true
  + restart             = "no"
  + rm                  = false
  + runtime             = (known after apply)
  + security_opts       = (known after apply)
  + shm_size            = (known after apply)
  + start               = true
  + stdin_open          = false
  + stop_signal         = (known after apply)
  + stop_timeout        = (known after apply)
  + tty                 = false
  + wait                = false
  + wait_timeout        = 60
}

```

```

+ wait                                = false
+ wait_timeout                        = 60

+ healthcheck (known after apply)

+ labels (known after apply)

+ networks_advanced {
+   aliases = []
+   driver_opts = {}
+   link_local_ips = []
+   name = "tf-classroom-net"
+   # {} unchanged attributes hidden
+ }

+ ports {
+   external = 8093
+   internal = 80
+   ip = "127.0.0.1"
+   protocol = "tcp"
+ }
}

# docker_container.web[3] will be created
+ resource "docker_container" "web" {
+   attach = false
+   bridge = (known after apply)
+   command = (known after apply)
+   container_logs = (known after apply)
+   container_read_refresh_timeout_milliseconds = 15000
+   entrypoint = (known after apply)
+   env = (known after apply)
+   exit_code = (known after apply)
+   hostname = (known after apply)
+   id = (known after apply)
+   image = "sha256:0985e772fb9f779e5fa898da85fca5d9c468e878eed43071645afe9d2c27094"
+   init = (known after apply)
+   ipc_mode = (known after apply)
+   log_driver = (known after apply)
+   logs = false
+   memory_reservation = 0
+   must_run = true
+   name = "tf-web-4"
+   network_data = (known after apply)
+   network_mode = "bridge"
+   platform = (known after apply)
+   read_only = false
+   remove_volumes = true
+   restart = "no"
+   rm = false
+   runtime = (known after apply)
+   security_opts = (known after apply)
+   size = (known after apply)
+   start = true
+   stdin_open = false
+   stop_signal = (known after apply)
+   stop_timeout = false
+   tty = false
+   wait = false
+   wait_timeout = 60
}

```

```

+ wait_timeout                        = 60

+ healthcheck (known after apply)

+ labels (known after apply)

+ networks_advanced {
+   aliases = []
+   driver_opts = {}
+   link_local_ips = []
+   name = "tf-classroom-net"
+   # {} unchanged attributes hidden
+ }

+ ports {
+   external = 8094
+   internal = 80
+   ip = "127.0.0.1"
+   protocol = "tcp"
+ }
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
- container_names = [
  # {} unchanged element hidden
  "tf-web-2",
  "tf-web-3",
  "tf-web-4",
]
- web_urls = [
  # {} unchanged element hidden
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]

```

Saved the plan to: initial.tfplan

To perform exactly these actions, run the following command to apply:

```
terraform apply "initial.tfplan"
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# docker_container.web[2] will be created
+ resource "docker_container" "web" {
+   attach = false
+   bridge = (known after apply)
+   command = (known after apply)
+   container_logs = (known after apply)
+   container_read_refresh_timeout_milliseconds = 15000
+   entrypoint = (known after apply)
+   env = (known after apply)
}

```

```

+ command = (known after apply)
+ container_logs = (known after apply)
+ container_read_refresh_timeout_milliseconds = 15000
+ entrypoint = (known after apply)
+ env = (known after apply)
+ exit_code = (known after apply)
+ hostname = (known after apply)
+ id = (known after apply)
+ image = "sha256:0986e772fb9f729e6fa8988da85fca5d9c468a878eed43071545af9d2e27d94"
+ init = (known after apply)
+ ipc_mode = (known after apply)
+ log_driver = (known after apply)
+ logs = false
+ memory_reservation = 0
+ must_run = true
+ name = "tf-web-3"
+ network_data = (known after apply)
+ network_mode = "bridge"
+ platform = (known after apply)
+ read_only = false
+ remove_volumes = true
+ restart = "no"
+ rm = false
+ runtime = (known after apply)
+ security_opts = (known after apply)
+ shm_size = (known after apply)
+ start = true
+ stdin_open = false
+ stop_signal = (known after apply)
+ stop_timeout = (known after apply)
+ tty = false
+ wait = false
+ wait_timeout = 60

+ healthcheck (known after apply)

+ labels (known after apply)

+ networks_advanced {
+   aliases = []
+   driver_opts = []
+   link_local_ips = []
+   name = "tf-classroom-net"
+   # (3 unchanged attributes hidden)
+ }

+ ports {
+   external = 8093
+   internal = 80
+   ip = "127.0.0.1"
+   protocol = "tcp"
+ }

# docker_container.web[3] will be created
+ resource "docker_container" "web" {
+   attach = false
+   bridge = (known after apply)
+   command = (known after apply)
+   container_logs = (known after apply)

```

```

+ container_logs = (known after apply)
+ container_read_refresh_timeout_milliseconds = 15000
+ entrypoint = (known after apply)
+ env = (known after apply)
+ exit_code = (known after apply)
+ hostname = (known after apply)
+ id = (known after apply)
+ image = "sha256:0986e772fb9f729e6fa8988da85fca5d9c468a878eed43071545af9d2e27d94"
+ init = (known after apply)
+ ipc_mode = (known after apply)
+ log_driver = (known after apply)
+ logs = false
+ memory_reservation = 0
+ must_run = true
+ name = "tf-web-4"
+ network_data = (known after apply)
+ network_mode = "bridge"
+ platform = (known after apply)
+ read_only = false
+ remove_volumes = true
+ restart = "no"
+ rm = false
+ runtime = (known after apply)
+ security_opts = (known after apply)
+ shm_size = (known after apply)
+ start = true
+ stdin_open = false
+ stop_signal = (known after apply)
+ stop_timeout = (known after apply)
+ tty = false
+ wait = false
+ wait_timeout = 60

+ healthcheck (known after apply)

+ labels (known after apply)

+ networks_advanced {
+   aliases = []
+   driver_opts = []
+   link_local_ips = []
+   name = "tf-classroom-net"
+   # (3 unchanged attributes hidden)
+ }

+ ports {
+   external = 8094
+   internal = 80
+   ip = "127.0.0.1"
+   protocol = "tcp"
+ }

}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
- container_names = [
  # (1 unchanged element hidden)
  "tf-web-2",
  "tf-web-3",

```

```

- "tf-web-3",
- "tf-web-4",
]
- web_urls = [
  # (id=6585c316f2a2852e277f47ec4e8465c7bd13f98c273d469dcbfb66f3cc53)
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]
docker_container.web[3]: Creating...
docker_container.web[2]: Creating...
docker_container.web[3]: Creation complete after 0s [id=6585c316f2a2852e277f47ec4e8465c7bd13f98c273d469dcbfb66f3cc53]
docker_container.web[2]: Creation complete after 0s [id=8635b6b6b4db16b902b38453e88b277ca39f10bd8979ae1dcb80a2ffc566c344]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```

Outputs:

```

container_names = [
  "tf-web-1",
  "tf-web-2",
  "tf-web-3",
  "tf-web-4",
]
web_urls = [
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]
anushkaberlia@Anushkas-MacBook-Air-2: ~ % terraform output
terraform state list
docker ps --filter name=tf-web
curl -fsS http://127.0.0.1:8091/
curl -fsS http://127.0.0.1:8092/
container_names = [
  "tf-web-1",
  "tf-web-2",
  "tf-web-3",
  "tf-web-4",
]
web_urls = [
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]
docker_container.web[0]
docker_container.web[1]
docker_container.web[2]
docker_container.web[3]
docker_image.web
docker_network.lab
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
8893b6b6b4db   8985e772fb9f   "/docker-entrypoint..." 3 seconds ago   Up 3 seconds   127.0.0.1:8093->80/tcp              tf-web-3
6585c316f2a2   8985e772fb9f   "/docker-entrypoint..." 3 seconds ago   Up 3 seconds   127.0.0.1:8094->80/tcp              tf-web-4
6585a952a0b4   8985e772fb9f   "/docker-entrypoint..." 10 hours ago    Up 10 hours    127.0.0.1:8092->80/tcp              tf-web-2
b3b8d3b9ce7e   8985e772fb9f   "/docker-entrypoint..." 10 hours ago    Up 10 hours    127.0.0.1:8091->80/tcp              tf-web-1
</DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 350px; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">F5.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
</DOCTYPE html>
<html>
<head>
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body { width: 350px; margin: 0 auto;
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</style>
</head>
<body>
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<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
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For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">F5.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
anushkaberlia@Anushkas-MacBook-Air-2: ~ % terraform N      for port in 8091 8092 8093 8094; do
  curl -fsS -o /dev/null -w "%{http_code}" "http://127.0.0.1:$port/"
done
8091 HTTP/200
8092 HTTP/200
8093 HTTP/200

```

```

</DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 350px; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">F5.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
anushkaberlia@Anushkas-MacBook-Air-2: ~ % terraform N      for port in 8091 8092 8093 8094; do
  curl -fsS -o /dev/null -w "%{http_code}" "http://127.0.0.1:$port/"
done
8091 HTTP/200
8092 HTTP/200
8093 HTTP/200

```

```

8094 HTTP/200
anushkaberlia@Anushkas-MacBook-Air-2 % terraform N cat > terraform.tfvars << 'EOF'
  replica_count = 2
  base_port     = 8091
  docker_host   = "unix:///Users/anushkaberlia/.docker/run/docker.sock"
EOF
heredoc> terraform plan -out=scale-down.tfplan
terraform show scale-down.tfplan
terraform apply scale-down.tfplan
terraform state list
heredoc>
anushkaberlia@Anushkas-MacBook-Air-2 % terraform % printf 'replica_count = %d\nbase_port = %d\ndocker_host = %s\n' > terraform.tfvars
anushkaberlia@Anushkas-MacBook-Air-2 % cat terraform.tfvars
replica_count = 2
base_port     = 8091
docker_host   = "unix:///Users/anushkaberlia/.docker/run/docker.sock"
anushkaberlia@Anushkas-MacBook-Air-2 % terraform N terraform plan -out=scale-down.tfplan
terraform show scale-down.tfplan
terraform apply scale-down.tfplan
terraform state list
docker_image.web: Refreshing state... [id=sha256:8986772fb97729e6fa9088da85fca5d9c46a078e0d3071545ef0d2c27094nginx:stable-alpine]
docker_network.lab: Refreshing state... [id=72b0d632ee28179c8a7b1835e27c43a849cfc0b339d8d3ea63a06f2c74372e]
docker_container.web[2]: Refreshing state... [id=8035b4eb4db16b902b30453e88b277c439f16b09099ae1db0ba2ffc5666344]
docker_container.web[0]: Refreshing state... [id=b3b6d3b9ce7e488128ade11862b7ecfba54fa76fb08ab6e23a60f2cc94f0fd]
docker_container.web[1]: Refreshing state... [id=808049c2a8a7fca8a82b0c0d465cb0ccdb34320d1097f1187fab2f6e222227f6]
docker_container.web[3]: Refreshing state... [id=6886c316f22803e277f47d7c4e8462c7b3d3f96c273846d0db7bf08f3ccc3]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  ~ destroy

Terraform will perform the following actions:

# docker_container.web[2] will be destroyed
# (because index [2] is out of range for count)
- resource "docker_container" "web" {
  ~ attach              = false -> null
  ~ command             = [
    ~ "nginx",
    ~ "-g",
    ~ "daemon off;",
  ] -> null
  ~ container_read_refresh_timeout_milliseconds = 15000 -> null
  ~ cpu_shares          = 0 -> null
  ~ dns                 = [ ] -> null
  ~ dns_opts            = [ ] -> null
  ~ dns_search          = [ ] -> null
  ~ endpoint            = [ ] -> null
  ~ /docker-entrypoint.sh,
  ] -> null
  ~ env                = [ ] -> null
  ~ group_add           = [ ] -> null
  ~ hostname            = "8035b4eb4db" -> null
  ~ id                  = "8035b4eb4db16b902b30453e88b277c439f16b09099ae1db0ba2ffc5666344" -> null
  ~ image               = "sha256:0985e772fb97729e6fa9088da85fca5d9c46a078e0d3071545ef0d2c27094" -> null
  ~ init               = false -> null
  ~ ipc_mode            = "private" -> null
  ~ log_driver          = "json-file" -> null
  ~ log_opts            = [ ] -> null
  ~ logs               = false -> null
}

```

```

- init                = false -> null
- ipc_mode            = "private" -> null
- log_driver          = "json-file" -> null
- log_opts            = [ ] -> null
- logs               = false -> null
- max_retry_count     = 0 -> null
- memory              = 0 -> null
- memory_reservation = 0 -> null
- memory_swap         = 0 -> null
- must_run            = true -> null
- name                = "tf-web-3" -> null
- network_data        = [ ]
- {
  ~ gateway            = "172.22.0.1"
  ~ global_ipv6_prefix_length = 0
  ~ ip_address         = "172.22.0.5"
  ~ ip_prefix_length   = 16
  ~ mac_address        = "e2:ec:c1:7:26:9d:a9"
  ~ network_name       = "tf-classroom-net"
  # (2 unchanged attributes hidden)
}
- } -> null
- network_mode        = "bridge" -> null
- platform            = "linux" -> null
- privileged           = false -> null
- publish_all_ports   = false -> null
- read_only           = false -> null
- remove_volumes      = true -> null
- restart             = "no" -> null
- rm                  = false -> null
- runtime             = "runc" -> null
- security_opts       = [ ] -> null
- shm_size            = 64 -> null
- start               = true -> null
- stdin_open          = false -> null
- stop_signal         = "SIGQUIT" -> null
- stop_timeout        = 0 -> null
- storage_opts        = [ ] -> null
- sysctl              = [ ] -> null
- tmpfs               = [ ] -> null
- tty                 = false -> null
- wait                = false -> null
- wait_timeout        = 60 -> null
- working_dir         = "/" -> null
  # (1 unchanged attribute hidden)
}

- networks_advanced {
  ~ aliases            = [ ] -> null
  ~ driver_opts        = [ ] -> null
  ~ gw_priority        = 0 -> null
  ~ link_local_ips     = [ ] -> null
  ~ name               = "tf-classroom-net" -> null
  # (3 unchanged attributes hidden)
}

- ports {
  ~ external = 8093 -> null
  ~ internal = 80 -> null
  ~ ip       = "172.0.0.1" -> null
  ~ protocol = "tcp" -> null
}

```

```

- protocol = "tcp" -> null
}
}

# docker_container.web[3] will be destroyed
# (because index [3] is out of range for count)
resource "docker_container" "web" {
  attach      = false -> null
  command     = [
    "nginx",
    "-g",
    "daemon off;",
  ]
  container_read_refresh_timeout_milliseconds = 15000 -> null
  cpu_shares  = 0 -> null
  dns         = [] -> null
  dns_opts   = [] -> null
  dns_search  = [] -> null
  entrypoint  = [
    "/docker-entrypoint.sh",
  ] -> null
  env         = [] -> null
  env_add     = [] -> null
  group_add   = ["6585c316f2a2"] -> null
  hostname    = "6585c316f2a2" -> null
  id          = "6585c316f2a2052c2f7470c4e0445c7b7d13f98c273d449d87bf66f3cc53" -> null
  image       = "sha256:098057727b9f779e5fa0980a05fca5d9c4d8a878edc3872545ef9d2e27d94" -> null
  init        = false -> null
  ipc_mode    = "private" -> null
  log_driver  = "json-file" -> null
  log_opts    = {} -> null
  logs        = false -> null
  max_retry_count = 0 -> null
  memory      = 0 -> null
  memory_reservation = 0 -> null
  memory_swap = 0 -> null
  must_run    = true -> null
  name        = "tf-web-4" -> null
  network_data = {
    gateway = "172.22.0.1"
    global_ipv6_prefix_length = 0
    ip_address = "172.22.0.4"
    ip_prefix_length = 16
    mac_address = "e2:08:e3:94:75:6a"
    network_name = "tf-classroom-net"
    # (1 unchanged attributes hidden)
  }
}

network_mode = "bridge" -> null
platform     = "linux" -> null
privileged   = false -> null
publish_all_ports = false -> null
read_only    = false -> null
remove_volumes = true -> null
restart      = "no" -> null
rm           = false -> null
runtime      = "runc" -> null
security_opts = [] -> null
shm_size     = 64 -> null
start        = true -> null

```

```

shm_size     = 64 -> null
start        = true -> null
stdin_open   = false -> null
stop_signal  = "SIGQUIT" -> null
stop_timeout = 0 -> null
storage_opts = {} -> null
sysctl       = {} -> null
tmpfs        = {} -> null
tty          = false -> null
wait         = false -> null
wait_timeout = 60 -> null
working_dir   = "/" -> null
# (1 unchanged attributes hidden)

networks_advanced {
  aliases = [] -> null
  driver_opts = {} -> null
  gw_priority = 0 -> null
  link_local_ips = [] -> null
  name = "tf-classroom-net" -> null
  # (1 unchanged attributes hidden)
}

ports {
  external = 8094 -> null
  internal = 80 -> null
  ip = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

```

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

```

- container_names = [
  # (1 unchanged element hidden)
  "tf-web-2",
  "tf-web-3",
  "tf-web-4",
]
- web_urls = [
  # (1 unchanged element hidden)
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]

```

Saved the plan to: scale-down.tfplan

To perform exactly these actions, run the following command to apply:
 terraform apply "scale-down.tfplan"

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

```

- destroy

```

Terraform will perform the following actions:

```

Terraform will perform the following actions:

# docker_container.web[2] will be destroyed
# (because index [2] is out of range for count)
- resource "docker_container" "web" {
  attach      = false -> null
  command     = [
    - "nginx",
    - "-g",
    - "daemon off;",
  ] -> null
  container_read_refresh_timeout_milliseconds = 15000 -> null
  cpu_shares  = 0 -> null
  dns         = [] -> null
  dns_opts   = [] -> null
  dns_search = [] -> null
  endpoint    = [
    - "/docker-entrypoint.sh",
  ] -> null
  env         = [] -> null
  group_add   = [] -> null
  hostname    = "8035b4eb4db" -> null
  id          = "8035b4eb4db166902b39453e88b277c439f16bd9989ae1ddb08e2ffc5666344" -> null
  image       = "sha256:0985e772fb9f729e6fa9908da85fca5d9c468e870eed43071545ef9d2e27d94" -> null
  init        = false -> null
  ipc_mode    = "private" -> null
  log_driver   = "json-file" -> null
  log_opts    = {} -> null
  logs        = false -> null
  max_retry_count = 0 -> null
  memory       = 0 -> null
  memory_reservation = 0 -> null
  memory_swap   = 0 -> null
  must_run      = true -> null
  name         = "tf-web-3" -> null
  network_data = {
    gateway          = "172.22.0.1"
    global_ipv6_prefix_length = 0
    ip_address       = "172.22.0.8"
    ip_prefix_length = 14
    mac_address      = "ee:c1:75:9d:a9"
    network_name     = "tf-classroom-net"
  }
  # (3 unchanged attributes hidden)
} -> null
- network_mode = "bridge" -> null
- platform     = "linux" -> null
- privileged   = false -> null
- publish_all_ports = false -> null
- read_only    = false -> null
- remove_volumes = true -> null
- restart      = "no" -> null
- rm           = false -> null
- runtime      = "runc" -> null
- security_opts = [] -> null
- shm_size     = 64 -> null
- start        = true -> null
- stdin_open   = false -> null
- stop_signal   = "SIGQUIT" -> null
- stop_timeout = 0 -> null
- storage_opts = {} -> null
- sysctl       = {} -> null
- tmpfs        = {} -> null
- tty          = false -> null
- wait         = false -> null
- wait_timeout = 60 -> null
- working_dir   = "/" -> null
  # (4 unchanged attributes hidden)
}

networks.advanced {
  aliases = [] -> null
  driver_opts = {} -> null
  gw_priority = 0 -> null
  link_local_ips = [] -> null
  name          = "tf-classroom-net" -> null
  # (3 unchanged attributes hidden)
}

ports {
  external = 8073 -> null
  internal = 80 -> null
  ip       = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

# docker_container.web[3] will be destroyed
# (because index [3] is out of range for count)
- resource "docker_container" "web" {
  attach      = false -> null
  command     = [
    - "nginx",
    - "-g",
    - "daemon off;",
  ] -> null
  container_read_refresh_timeout_milliseconds = 15000 -> null
  cpu_shares  = 0 -> null
  dns         = [] -> null
  dns_opts   = [] -> null
  dns_search = [] -> null
  endpoint    = [
    - "/docker-entrypoint.sh",
  ] -> null
  env         = [] -> null
  group_add   = [] -> null
  hostname    = "6585c316f2a2" -> null
  id          = "6585c316f2a2052c2f7f4d7ec4e8445cfb7d13f98c273d46vdbfbf68f3cc63" -> null
  image       = "sha256:0985e772fb9f729e6fa9908da85fca5d9c468e870eed43071545ef9d2e27d94" -> null
  init        = false -> null
  ipc_mode    = "private" -> null
  log_driver   = "json-file" -> null
  log_opts    = {} -> null
  logs        = false -> null
  max_retry_count = 0 -> null
  memory       = 0 -> null
  memory_reservation = 0 -> null

```

```

- start        = true -> null
- stdin_open   = false -> null
- stop_signal   = "SIGQUIT" -> null
- stop_timeout = 0 -> null
- storage_opts = {} -> null
- sysctl       = {} -> null
- tmpfs        = {} -> null
- tty          = false -> null
- wait         = false -> null
- wait_timeout = 60 -> null
- working_dir   = "/" -> null
  # (4 unchanged attributes hidden)
}

networks.advanced {
  aliases = [] -> null
  driver_opts = {} -> null
  gw_priority = 0 -> null
  link_local_ips = [] -> null
  name          = "tf-classroom-net" -> null
  # (3 unchanged attributes hidden)
}

ports {
  external = 8073 -> null
  internal = 80 -> null
  ip       = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

# docker_container.web[3] will be destroyed
# (because index [3] is out of range for count)
- resource "docker_container" "web" {
  attach      = false -> null
  command     = [
    - "nginx",
    - "-g",
    - "daemon off;",
  ] -> null
  container_read_refresh_timeout_milliseconds = 15000 -> null
  cpu_shares  = 0 -> null
  dns         = [] -> null
  dns_opts   = [] -> null
  dns_search = [] -> null
  endpoint    = [
    - "/docker-entrypoint.sh",
  ] -> null
  env         = [] -> null
  group_add   = [] -> null
  hostname    = "6585c316f2a2" -> null
  id          = "6585c316f2a2052c2f7f4d7ec4e8445cfb7d13f98c273d46vdbfbf68f3cc63" -> null
  image       = "sha256:0985e772fb9f729e6fa9908da85fca5d9c468e870eed43071545ef9d2e27d94" -> null
  init        = false -> null
  ipc_mode    = "private" -> null
  log_driver   = "json-file" -> null
  log_opts    = {} -> null
  logs        = false -> null
  max_retry_count = 0 -> null
  memory       = 0 -> null
  memory_reservation = 0 -> null

```

```

log_opts = {} -> null
logs = false -> null
max_retry_count = 5 -> null
memory = 0 -> null
memory_reservation = 0 -> null
memory_swap = 0 -> null
must_run = true -> null
name = "tf-web-4" -> null
network_data = {
  gateway = "172.22.0.1"
  global_ipv6_prefix_length = 0
  ip_address = "172.22.0.4"
  ip_prefix_length = 16
  mac_address = "e2:c8:e3:94:75:8a"
  network_name = "tf-classroom-net"
} # (2 unchanged attributes hidden)
} -> null
network_mode = "bridge" -> null
platform = "linux" -> null
privileged = false -> null
publish_all_ports = false -> null
read_only = false -> null
remove_volumes = true -> null
restart = "no" -> null
rm = false -> null
runtime = "runc" -> null
security_opts = [] -> null
shm_size = 64 -> null
start = true -> null
stdin_open = false -> null
stop_signal = "SIGQUIT" -> null
stop_timeout = 0 -> null
storage_opts = {} -> null
sysctls = {} -> null
tmpfs = {} -> null
tty = false -> null
wait = false -> null
wait_timeout = 60 -> null
working_dir = "/" -> null
} # (2 unchanged attributes hidden)

networks_advanced {
  classes = [] -> null
  driver_opts = {} -> null
  gw_priority = 0 -> null
  link_local_ips = [] -> null
  name = "tf-classroom-net" -> null
} # (2 unchanged attributes hidden)

}

ports {
  external = 8094 -> null
  internal = 88 -> null
  ip = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

```

```

}
}

ports {
  external = 8094 -> null
  internal = 88 -> null
  ip = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:
- container_names = [
  "tf-web-2",
  "tf-web-3",
  "tf-web-4",
]
- web_urls = [
  "http://127.0.0.1:8092",
  "http://127.0.0.1:8093",
  "http://127.0.0.1:8094",
]
}

docker_container.web[2]: Destroying... [id=0038b5b5eb4d31ab92b30453e88b277c439f14bd9999ae1dd508a2ffc5666364]
docker_container.web[3]: Destroying... [id=60886c316f2a2852e277f47d7ec4e8a6dcfb7d13f98c273d469ddbf05f3cc53]
docker_container.web[2]: Destruction complete after 0s
docker_container.web[3]: Destruction complete after 0s

Apply complete! Resources: 0 added, 0 changed, 2 destroyed.

Outputs:
container_names = [
  "tf-web-1",
  "tf-web-2",
]
web_urls = [
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
]
}
docker_container.web[0]
docker_container.web[1]
docker_image.web
docker_network.lab
amushkaber11@Amushkas-MacBook-Air:~$ terraform plan -destroy -out=destroy.tfplan
terraform show destroy.tfplan
terraform apply destroy.tfplan
terraform state list
docker ps --filter name=tf-web
docker_network.lab: Refreshing state... [id=73b0dd3eeea8179c8a7b1935e27c43a4d83cfc00397d0d30a56326f2e743f2e]
docker_image.web: Refreshing state... [id=a226d1e9867727b9f72e4d1d998a5e8fca89c44da878e0d4387154dafa9d1e27094a9mra:stable-alpine]
docker_container.web[1]: Refreshing state... [id=68664952a6bafca8a2b2c05d4d5c8b0acc352A325d1b9711870ab2f6ed222f76]
docker_container.web[0]: Refreshing state... [id=b3b8d3b9ce7ee88128ada711862b7ecfba54fa76fb88ab6a3a48f2cc094f09d]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
- destroy

Terraform will perform the following actions:

```

Terraform will perform the following actions:

```
# docker_container.web[0] will be destroyed
resource "docker_container" "web" {
  attach      = false
  command     = ["nginx"]
  dns         = ["g"]
  dns_search  = ["daemon off;"]
  entrypoint  = ["/docker-entrypoint.sh"]
  group_add   = []
  hostname    = "b388d3b9ce7e"
  id         = "b388d3b9ce7e88128ada7118a2b7ecfba54fa76f088ab6a3a48f2cec94fd9d"
  image       = "sha256:0985e772fb9f729e5fa9908da85fca5d9c468a870eed43071545efa9d2e27d94"
  init        = false
  ipc_mode    = "private"
  log_driver  = "json-file"
  log_opts    = {}
  logs        = false
  max_retry_count = 0
  memory      = 0
  memory_reservation = 0
  memory_swap = 0
  must_run    = true
  name        = "tf-web-1"
  network_data {
    gateway      = "172.22.0.1"
    global_ipv6_prefix_length = 0
    ip_address   = "172.22.0.3"
    ip_prefix_length = 16
    mac_address  = "62:fe:a7:5c:6a:f0"
    network_name = "tf-classroom-net"
  }
}

network_mode = "bridge"
platform     = "linux"
privileged   = false
publish_all_ports = false
read_only    = false
remove_volumes = true
restart      = "no"
rm           = false
runtime      = "runc"
security_opts = []
shm_size     = 64
start        = true
stdin_open   = false
stop_signal  = "SIGQUIT"
```

```
tmpfs      = false
tty         = false
wait        = false
wait_timeout = 60
working_dir = "/"

networks_advanced {
  aliases      = []
  driver_opts  = {}
  gw_priority   = 0
  link_local_ips = []
  name         = "tf-classroom-net"
}

ports {
  external = 8091
  internal = 80
  ip       = "172.0.0.1"
  protocol = "tcp"
}

# docker_container.web[1] will be destroyed
resource "docker_container" "web" {
  attach      = false
  command     = ["nginx"]
  dns         = ["g"]
  dns_search  = ["daemon off;"]
  entrypoint  = ["/docker-entrypoint.sh"]
  group_add   = []
  hostname    = "86854952abb4"
  id         = "86854952abb4fca807bcd4dd6cb884c49624326d1b9711876ab2f6ed222f76"
  image       = "sha256:0985e772fb9f729e5fa9908da85fca5d9c468a870eed43071545efa9d2e27d94"
  init        = false
  ipc_mode    = "private"
  log_driver  = "json-file"
  log_opts    = {}
  logs        = false
  max_retry_count = 0
  memory      = 0
  memory_reservation = 0
  memory_swap = 0
  must_run    = true
  name        = "tf-web-2"
  network_data {
    gateway      = "172.22.0.1"
    global_ipv6_prefix_length = 0
  }
}
```

```

- gateway           = "172.22.0.1"
- global_ipv6_prefix_length = 0
- ip_address        = "172.22.0.2"
- ip_prefix_length  = 16
- mac_address       = "2e:41:db:ba:2e:2e"
- network_name      = "tf-classroom-net"
  # (1 unchanged attributes hidden)
}

network_mode = "bridge" -> null
platform     = "linux"  -> null
privileged   = false    -> null
publish_all_ports = false -> null
read_only    = false    -> null
remove_volumes = true   -> null
restart      = "no"     -> null
rm           = false    -> null
runtime      = "runc"   -> null
security_opts = []      -> null
shm_size     = 64       -> null
start        = true     -> null
stdin_open   = false    -> null
stop_signal  = "SIGQUIT" -> null
stop_timeout = 0        -> null
storage_opts = {}       -> null
sysctl       = {}       -> null
tmpfs        = {}       -> null
tty          = false    -> null
wait         = false    -> null
wait_timeout = 60       -> null
working_dir  = "/"       -> null
# (1 unchanged attributes hidden)

networks_advanced {
  aliases = [] -> null
  driver_opts = {} -> null
  gw_priority = 0 -> null
  link_local_ips = [] -> null
  name = "tf-classroom-net" -> null
  # (1 unchanged attributes hidden)
}

ports {
  external = 8092 -> null
  internal = 80   -> null
  ip       = "127.0.0.1" -> null
  protocol = "tcp" -> null
}
}

# docker_image.web will be destroyed
resource "docker_image" "web" {
  id = "sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
  image_id = "sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
  keep_locally = true -> null
  name = "nginx:stable-alpine" -> null
  repo_digest = "nginx@sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
}

# docker_network.lab will be destroyed

```

```

# docker_image.web will be destroyed
resource "docker_image" "web" {
  id = "sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
  image_id = "sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
  keep_locally = true -> null
  name = "nginx:stable-alpine" -> null
  repo_digest = "nginx@sha256:098be772fb9f729e6fa0980da05fca5d9c468e870eed43071545afa9d2e27d94" -> null
}

# docker_network.lab will be destroyed
resource "docker_network" "lab" {
  attachable = false -> null
  driver     = "bridge" -> null
  id        = "73acd3eeea81709c8afb1835e27c43a68d3cfc5b3398d3ea562d5f2e74372e" -> null
  ingress    = false -> null
  internal   = false -> null
  ipam_driver = "default" -> null
  ipam_options = {} -> null
  ipv6       = false -> null
  name       = "tf-classroom-net" -> null
  options     = {
    "com.docker.network.enable_ipv4" = "true"
    "com.docker.network.enable_ipv6" = "false"
  } -> null
  scope      = "local" -> null

  ipam_config {
    aux_address = {} -> null
    gateway     = "172.22.0.1" -> null
    subnet      = "172.22.0.0/16" -> null
  }
  # (1 unchanged attributes hidden)
}

Plan: 0 to add, 0 to change, 4 to destroy.

Changes to Outputs:
- container_names = [
  "tf-web-1",
  "tf-web-2",
] -> null
- web_urls = [
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
] -> null

```

Saved the plan to: destroy.tfplan

To perform exactly these actions, run the following command to apply:
 terraform apply "destroy.tfplan"

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
 destroy

Terraform will perform the following actions:

docker_container.web[0] will be destroyed

```

- resource "docker_container" "web" {
-   attach                = false -> null
-   command               = [
-     "nginx",
-     "-g",
-     "daemon off;",
-   ] -> null
-   container_read_refresh_timeout_milliseconds = 15000 -> null
-   cpu_shares             = 0 -> null
-   dns                   = [] -> null
-   dns_opts              = [] -> null
-   dns_search            = [] -> null
-   entrypoint            = [
-     "/docker-entrypoint.sh",
-   ] -> null
-   env                   = [] -> null
-   group_add             = [] -> null
-   hostname              = "b3b8d3b9ce7e" -> null
-   id                    = "b3b8d3b9ce7e" -> null
-   image                 = "sha256:0986e772fb9f729e6fa9988da85fca5d9c468e878eed43671545efa9d2e27d94" -> null
-   init                  = false -> null
-   ipc_mode              = "private" -> null
-   log_driver            = "json-file" -> null
-   log_opts              = {} -> null
-   logs                  = false -> null
-   max_retry_count       = 0 -> null
-   memory                = 0 -> null
-   memory_reservation    = 0 -> null
-   memory_swap           = 0 -> null
-   must_run              = true -> null
-   name                  = "tf-web-1" -> null
-   network_data          = {
-     gateway              = "172.22.0.1"
-     global_ipv6_prefix_length = 0
-     ip_address           = "172.22.0.3"
-     ip_prefix_length     = 16
-     mac_address          = "62:fe:a7:5c:da:f0"
-     network_name         = "tf-classroom-net"
-   } # (2 unchanged attributes hidden)
- } -> null
- network_mode            = "bridge" -> null
- platform               = "linux" -> null
- privileged              = false -> null
- publish_all_ports       = false -> null
- read_only               = false -> null
- remove_volumes         = true -> null
- restart                 = "no" -> null
- rm                      = false -> null
- runtime                 = "runc" -> null
- security_opts           = [] -> null
- shm_size                = 64 -> null
- start                   = true -> null
- stdin_open             = false -> null
- stop_signal             = "SIGQUIT" -> null
- stop_timeout            = 0 -> null
- storage_opts            = {} -> null
- sysctls                 = {} -> null
- tmpfs                  = {} -> null

```

```

- networks_advanced {
-   aliases               = [] -> null
-   driver_opts           = {} -> null
-   gw_priority            = 0 -> null
-   link_local_ips        = [] -> null
-   name                  = "tf-classroom-net" -> null
- } # (3 unchanged attributes hidden)
- }
- ports {
-   external = 8091 -> null
-   internal = 80 -> null
-   ip       = "172.0.0.1" -> null
-   protocol = "tcp" -> null
- }
- # docker_container.web[1] will be destroyed
- resource "docker_container" "web" {
-   attach                = false -> null
-   command               = [
-     "nginx",
-     "-g",
-     "daemon off;",
-   ] -> null
-   container_read_refresh_timeout_milliseconds = 15000 -> null
-   cpu_shares             = 0 -> null
-   dns                   = [] -> null
-   dns_opts              = [] -> null
-   dns_search            = [] -> null
-   entrypoint            = [
-     "/docker-entrypoint.sh",
-   ] -> null
-   env                   = [] -> null
-   group_add             = [] -> null
-   hostname              = "85864952a6b4" -> null
-   id                    = "85864952a6b4" -> null
-   image                 = "sha256:0986e772fb9f729e6fa9988da85fca5d9c468e878eed43671545efa9d2e27d94" -> null
-   init                  = false -> null
-   ipc_mode              = "private" -> null
-   log_driver            = "json-file" -> null
-   log_opts              = {} -> null
-   logs                  = false -> null
-   max_retry_count       = 0 -> null
-   memory                = 0 -> null
-   memory_reservation    = 0 -> null
-   memory_swap           = 0 -> null
-   must_run              = true -> null
-   name                  = "tf-web-2" -> null
-   network_data          = {
-     gateway              = "172.22.0.1"
-     global_ipv6_prefix_length = 0
-     ip_address           = "172.22.0.2"
-     ip_prefix_length     = 16
-     mac_address          = "2e:41:db:be:2a:2a"
-     network_name         = "tf-classroom-net"
-   } # (2 unchanged attributes hidden)
- },

```

```

- network_name = "tf-classroom-net"
# if unchanged attribute hidden)
},
] => null
- network_mode = "bridge" => null
- platform = "linux" => null
- privileged = false => null
- publish_all_ports = false => null
- read_only = false => null
- remove_volumes = true => null
- restart = "no" => null
- rm = false => null
- runtime = "runc" => null
- security_opts = [] => null
- shm_size = 64 => null
- start = true => null
- stdin_open = false => null
- stop_signal = "SIGQUIT" => null
- stop_timeout = 0 => null
- storage_opts = {} => null
- sysctls = {} => null
- tmpfs = false => null
- wait_timeout = false => null
- working_dir = "" => null
# if unchanged attribute hidden)

- networks_advanced {
- aliases = [] => null
- driver_opts = {} => null
- gw_priority = 0 => null
- link_local_ips = [] => null
- name = "tf-classroom-net" => null
# if unchanged attribute hidden)
}

- ports {
- external = 8092 => null
- internal = 80 => null
- ip = "127.0.0.1" => null
- protocol = "tcp" => null
}

# docker_image.web will be destroyed
- resource "docker_image" "web" {
- id = "sha256:898a7721b9f7729e6fa0980da05fca5d9c468e870eed43071545afa9d2a27094" => null
- image_id = "sha256:898a7721b9f7729e6fa0980da05fca5d9c468e870eed43071545afa9d2a27094" => null
- keep_locally = true => null
- name = "nginx:stable-alpine" => null
- repo_digest = "nginx@sha256:898a7721b9f7729e6fa0980da05fca5d9c468e870eed43071545afa9d2a27094" => null
}

# docker_network.lab will be destroyed
- resource "docker_network" "lab" {
- attachable = false => null
- driver = "bridge" => null
- id = "73bcd03aee817b9c8afb1835e27c43a68d3cfc5b339d8d3ea562d5f2e743f2e" => null
- ingress = false => null
- ipam_driver = "default" => null
- ipam_options = {} => null
- ipv6 = false => null
- name = "tf-classroom-net" => null
- options = {
- "com.docker.network.enable_ipv4" = "true"
- "com.docker.network.enable_ipv6" = "false"
} => null
- scope = "local" => null
- ipam_config {
- aux_address = {} => null
- gateway = "172.22.0.1" => null
- subnet = "172.22.0.0/16" => null
# if unchanged attribute hidden)
}
}

```

```

- repo_digest = "nginx@sha256:898a7721b9f7729e6fa0980da05fca5d9c468e870eed43071545afa9d2a27094" => null
}

# docker_network.lab will be destroyed
- resource "docker_network" "lab" {
- attachable = false => null
- driver = "bridge" => null
- id = "73bcd03aee817b9c8afb1835e27c43a68d3cfc5b339d8d3ea562d5f2e743f2e" => null
- ingress = false => null
- ipam_driver = "default" => null
- ipam_options = {} => null
- ipv6 = false => null
- name = "tf-classroom-net" => null
- options = {
- "com.docker.network.enable_ipv4" = "true"
- "com.docker.network.enable_ipv6" = "false"
} => null
- scope = "local" => null
- ipam_config {
- aux_address = {} => null
- gateway = "172.22.0.1" => null
- subnet = "172.22.0.0/16" => null
# if unchanged attribute hidden)
}
}

Plan: 0 to add, 0 to change, 4 to destroy.

Changes to Outputs:
- container_names = [
- "tf-web-1",
- "tf-web-2",
] => null
- web_urls = [
- "http://127.0.0.1:8091",
- "http://127.0.0.1:8092",
] => null
docker_container.web[1]: Destruction complete after 0s
docker_container.web[0]: Destruction complete after 0s
docker_image.web: Destruction complete after 0s
docker_network.lab: Destruction complete after 0s
docker_image.web: Destruction complete after 2s
docker_network.lab: Destruction complete after 2s

Apply complete! Resources: 0 added, 0 changed, 4 destroyed.
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
anushkaberila@Anushka-MacBook-Air-2 % terraform status
terraform show initial.tfplan
terraform apply initial.tfplan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

```

```

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
enushkber10a8mushkas-MacBook-Air-2 62_terraform % terraform plan -out=initial.tfplan
terraform show initial.tfplan
terraform apply initial.tfplan

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# docker_container.web[0] will be created
+ resource "docker_container" "web" {
  + attach           = false
  + bridge           = (known after apply)
  + command          = (known after apply)
  + container_logs   = (known after apply)
  + container_read_refresh_timeout_milliseconds = 10000
  + endpoint         = (known after apply)
  + env              = (known after apply)
  + exit_code        = (known after apply)
  + hostname         = (known after apply)
  + id               = (known after apply)
  + image            = (known after apply)
  + init             = (known after apply)
  + ipc_mode         = (known after apply)
  + log_driver       = (known after apply)
  + logs             = false
  + memory_reservation = 0
  + must_run         = true
  + name             = "tf-web-1"
  + network_data     = (known after apply)
  + network_mode     = "bridge"
  + platform         = (known after apply)
  + read_only        = false
  + remove_volumes   = true
  + restart          = "no"
  + rm               = false
  + runtime          = (known after apply)
  + security_opts    = (known after apply)
  + shm_size         = (known after apply)
  + start            = true
  + stdin_open       = false
  + stop_signal      = (known after apply)
  + stop_timeout     = (known after apply)
  + tty              = false
  + wait             = false
  + wait_timeout     = 60

  + healthcheck (known after apply)

  + labels (known after apply)

  + networks_advanced {
    + aliases = []
    + driver_opts = {}
    + link_local_ips = []
    + name = "tf-classroom-net"
    # {} unchanged attributes hidden
  }
}

```

```

# {} unchanged attributes hidden
}

+ ports {
  + external = 8091
  + internal = 80
  + ip       = "127.0.0.1"
  + protocol = "tcp"
}

# docker_container.web[1] will be created
+ resource "docker_container" "web" {
  + attach           = false
  + bridge           = (known after apply)
  + command          = (known after apply)
  + container_logs   = (known after apply)
  + container_read_refresh_timeout_milliseconds = 10000
  + endpoint         = (known after apply)
  + env              = (known after apply)
  + exit_code        = (known after apply)
  + hostname         = (known after apply)
  + id               = (known after apply)
  + image            = (known after apply)
  + init             = (known after apply)
  + ipc_mode         = (known after apply)
  + log_driver       = (known after apply)
  + logs             = false
  + memory_reservation = 0
  + must_run         = true
  + name             = "tf-web-2"
  + network_data     = (known after apply)
  + network_mode     = "bridge"
  + platform         = (known after apply)
  + read_only        = false
  + remove_volumes   = true
  + restart          = "no"
  + rm               = false
  + runtime          = (known after apply)
  + security_opts    = (known after apply)
  + shm_size         = (known after apply)
  + start            = true
  + stdin_open       = false
  + stop_signal      = (known after apply)
  + stop_timeout     = (known after apply)
  + tty              = false
  + wait             = false
  + wait_timeout     = 60

  + healthcheck (known after apply)

  + labels (known after apply)

  + networks_advanced {
    + aliases = []
    + driver_opts = {}
    + link_local_ips = []
    + name = "tf-classroom-net"
    # {} unchanged attributes hidden
  }
}

```

```

# (3 unchanged attributes hidden)
}

+ ports {
  + external = 8092
  + internal = 88
  + ip       = "127.0.0.1"
  + protocol = "tcp"
}
}

# docker_image.web will be created
+ resource "docker_image" "web" {
  + id           = (known after apply)
  + image_id     = (known after apply)
  + keep_locally = true
  + name        = "nginx:stable-alpine"
  + repo_digest = (known after apply)
}

# docker_network.lab will be created
+ resource "docker_network" "lab" {
  + driver   = (known after apply)
  + id       = (known after apply)
  + internal = (known after apply)
  + ipam_driver = "default"
  + name      = "tf-classroom-net"
  + options   = (known after apply)
  + scope     = (known after apply)

  + ipam_config (known after apply)
}

Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ container_names = [
  - "tf-web-1",
  - "tf-web-2",
]
+ web_urls = [
  - "http://127.0.0.1:8091",
  - "http://127.0.0.1:8092",
]

Saved the plan to: initial.tfplan

To perform exactly these actions, run the following command to apply:
    terraform apply "initial.tfplan"

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# docker_container.web[0] will be created
+ resource "docker_container" "web" {

```

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# docker_container.web[0] will be created
+ resource "docker_container" "web" {
  + attach                = false
  + bridge                = (known after apply)
  + command               = (known after apply)
  + container_logs        = (known after apply)
  + container_read_refresh_timeout_milliseconds = 10000
  + entrypoint             = (known after apply)
  + env                   = (known after apply)
  + exit_code              = (known after apply)
  + hostname               = (known after apply)
  + id                    = (known after apply)
  + image                  = (known after apply)
  + init                   = (known after apply)
  + ipc_mode               = (known after apply)
  + log_driver             = (known after apply)
  + logs                   = false
  + memory_reservation     = 0
  + must_run               = true
  + name                   = "tf-web-1"
  + network_data           = (known after apply)
  + network_mode           = "bridge"
  + platform               = (known after apply)
  + read_only              = false
  + remove_volumes        = true
  + restart                = "no"
  + rm                     = false
  + runtime                 = (known after apply)
  + security_opts           = (known after apply)
  + shm_size                = (known after apply)
  + start                  = true
  + stdin_open             = false
  + stop_signal             = (known after apply)
  + stop_timeout           = (known after apply)
  + tty                    = false
  + wait                   = false
  + wait_timeout           = 60

  + healthcheck (known after apply)

  + labels (known after apply)

  + networks_advanced {
    + aliases = []
    + driver_opts = []
    + link_local_ips = []
    + name      = "tf-classroom-net"
  }
# (3 unchanged attributes hidden)
}

+ ports {
  + external = 8091
  + internal = 88
}
}

```

```

    ports {
      + external = 8091
      + internal = 88
      + ip       = "127.0.0.1"
      + protocol = "tcp"
    }
  }

# docker_container.web[1] will be created
+ resource "docker_container" "web" {
  + attach      = false
  + bridge      = (known after apply)
  + command     = (known after apply)
  + container_logs = (known after apply)
  + container_read_refresh_timeout_milliseconds = 15000
  + entrypoint   = (known after apply)
  + env         = (known after apply)
  + exit_code    = (known after apply)
  + hostname     = (known after apply)
  + id          = (known after apply)
  + image        = (known after apply)
  + init         = (known after apply)
  + ipc_mode     = (known after apply)
  + log_driver   = (known after apply)
  + logs        = false
  + memory_reservation = 0
  + must_run     = true
  + name         = "tf-web-2"
  + network_data = (known after apply)
  + network_mode = "bridge"
  + platform     = (known after apply)
  + read_only    = false
  + remove_volumes = true
  + restart      = "no"
  + rm           = false
  + runtime      = (known after apply)
  + security_opts = (known after apply)
  + shm_size     = (known after apply)
  + start        = true
  + stdin_open   = false
  + stop_signal  = (known after apply)
  + stop_timeout = (known after apply)
  + tty          = false
  + wait         = false
  + wait_timeout = 60

  + healthcheck (known after apply)

  + labels (known after apply)

  + networks_advanced {
    + aliases = []
    + driver_opts = []
    + link_local_ips = []
    + name = "tf-classroom-net"
  } # {} unchanged attributes hidden
}

+ ports {
  + external = 8092

```

```

    + internal = 88
    + ip       = "127.0.0.1"
    + protocol = "tcp"
  }
}

# docker_container.web[1] will be created
+ resource "docker_container" "web" {
  + attach      = false
  + bridge      = (known after apply)
  + command     = (known after apply)
  + container_logs = (known after apply)
  + container_read_refresh_timeout_milliseconds = 15000
  + entrypoint   = (known after apply)
  + env         = (known after apply)
  + exit_code    = (known after apply)
  + hostname     = (known after apply)
  + id          = (known after apply)
  + image        = (known after apply)
  + init         = (known after apply)
  + ipc_mode     = (known after apply)
  + log_driver   = (known after apply)
  + logs        = false
  + memory_reservation = 0
  + must_run     = true
  + name         = "tf-web-2"
  + network_data = (known after apply)
  + network_mode = "bridge"
  + platform     = (known after apply)
  + read_only    = false
  + remove_volumes = true
  + restart      = "no"
  + rm           = false
  + runtime      = (known after apply)
  + security_opts = (known after apply)
  + shm_size     = (known after apply)
  + start        = true
  + stdin_open   = false
  + stop_signal  = (known after apply)
  + stop_timeout = (known after apply)
  + tty          = false
  + wait         = false
  + wait_timeout = 60

  + healthcheck (known after apply)

  + labels (known after apply)

  + networks_advanced {
    + aliases = []
    + driver_opts = []
    + link_local_ips = []
    + name = "tf-classroom-net"
  } # {} unchanged attributes hidden
}

+ ports {
  + external = 8092
  + internal = 88
  + ip       = "127.0.0.1"

```

```

    + ip      = "127.0.0.1"
    + protocol = "tcp"
  }
}

# docker_image.web will be created
+ resource "docker_image" "web" {
  + id      = (known after apply)
  + image_id = (known after apply)
  + keep_locally = true
  + name     = "nginx:stable-alpine"
  + repo_digest = (known after apply)
}

# docker_network.lab will be created
+ resource "docker_network" "lab" {
  + driver      = (known after apply)
  + id          = (known after apply)
  + internal    = (known after apply)
  + ipam_driver = "default"
  + name        = "tf-classroom-net"
  + options     = (known after apply)
  + scope       = (known after apply)

  + ipam_config (known after apply)
}

Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ container_names = [
  + "tf-web-1",
  + "tf-web-2",
]
+ web_urls       = [
  + "http://127.0.0.1:8091",
  + "http://127.0.0.1:8092",
]
docker_network.lab: Creating...
docker_image.web: Creating...
docker_image.web: Creation complete after 0s [id=sha256:0985e7721b9f729e6fa0988da05fca5d9c440e870eed43071546af9d3e27d94rginx:stable-alpine]
docker_network.lab: Creation complete after 2s [id=241ae30206ae9d09a16a03eccf31b9d8da2a6d983e5ec94e986b7b5ae9d2b01c]
docker_container.web[0]: Creating...
docker_container.web[1]: Creating...
docker_container.web[1]: Creation complete after 0s [id=7401ef92260732217625d4ee34e9db757799bf113fadfe669a130cb26f37b09]
docker_container.web[0]: Creation complete after 0s [id=a3435e84d1fa1fec3e3dcfdab47c0d6cdc91610d5e154238b0824ac7beede]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:
container_names = [
  "tf-web-1",
  "tf-web-2",
]
web_urls = [
  "http://127.0.0.1:8091",
  "http://127.0.0.1:8092",
]

```

anushkaberila@Anushkas-MacBook-Air-2: ~ % terraform %